Test Plan — Flipkart QA Project

Test Plan ID: FLPK-TP-001 **Application:** flipkart.com (guest flows) **Prepared by:** Nishank
Companion project: SauceDemo QA Project

1. Objective

Validate the core guest shopping journey on flipkart.com through manual and automated testing, and demonstrate a complete QA process — planning, design, execution, defect management, automation, and CI/CD.

2. Scope

In scope: homepage, global search, sorting, filters, pagination, product detail page, and cart — all as a guest (non-logged-in) user. Full detail in *01_Requirements.md*.

Out of scope: OTP-verified login/checkout, real payment gateway integration, Flipkart Seller Hub flows, and the native mobile app (this project targets the website only).

3. Test Strategy

Type	Approach
Functional	Manual test cases + Selenium automation
API	REST Assured + Postman (DummyJSON — see API module README)
Regression	Automated suite re-run via Jenkins on every commit
Cross-browser (sample)	Chrome (primary), Firefox (smoke subset) via WebDriverManager
Exploratory	Ad-hoc session on search + filters, logged in the Defect Report

4. Test Environment

Item	Detail
Application	https://www.flipkart.com (production)
Browsers	Chrome (latest), Firefox (latest)
Automation stack	Java 11+, Selenium WebDriver, TestNG, Maven, Log4j, ExtentReports
API stack	REST Assured, Postman / Newman
CI	Jenkins (declarative pipeline)
OS	Windows / Linux / macOS (cross-platform via WebDriverManager)

5. Entry Criteria

- Requirements document reviewed and finalized

- Test environment (browsers, drivers, network) available
- Flipkart.com reachable and rendering normally

6. Exit Criteria

- 100% of planned test cases executed
- No open Critical/High severity defects in scope areas
- Automated regression suite passing on CI

7. Test Deliverables

- Requirements document
- Test plan (this document)
- Test scenarios and detailed test cases (Excel)
- Execution report (Excel)
- Defect report (Excel, with screenshot references)
- Automated Selenium/TestNG suite (with Extent HTML report + Log4j logs)
- Automated API suite + Postman collection
- Jenkins pipeline
- Final project report and presentation deck

8. Roles & Responsibilities

Role	Responsibility
QA Engineer (this project)	Test design, manual execution, automation, defect logging, CI setup

9. Risks & Mitigation

Risk	Mitigation
Flipkart UI/DOM changes frequently (live production site)	Page Object Model isolates locators to one place per page; explicit waits over hard sleeps
Prices/stock change constantly	Assertions target relative behavior (e.g. “sorted ascending”) not fixed values
No test/staging environment available	All tests run read-only guest actions; no destructive/purchase actions performed
OTP-gated login blocks authenticated testing	Explicitly scoped out; documented as a known limitation, not silently skipped

10. Schedule (self-paced portfolio project)

Phase	Deliverable
1	Requirements + Test Plan
2	Manual test design (scenarios + cases)
3	Manual execution + defect logging
4	Selenium/TestNG automation (POM, Extent, Log4j, config)
5	API automation (REST Assured + Postman)
6	CI/CD (Jenkins) integration
7	Final report, presentation, and GitHub upload